

TRANSCRIPT

Let's start with a dirty word shall we? Pandemic. As we transition out of the last pandemic, we're entering a new pandemic of addiction that might be the most pervasive addiction in human history.

The scary part is, the world has yet to even acknowledge that this addiction, despite it being all around us and fuelling our current mental health crisis. But today we turn the tide on this new addiction.

What if I were to tell you that in 2007, a new addictive drug came out and it took the world by storm. In less than 15 years, over half of the world's population was using this drug daily. It was causing a rise in depression, anxiety and suicide in all age ranges, but especially in young girls. And it was accounting for 3000 traffic deaths per year in the United States from impaired driving.

If this were happening, wouldn't there be an outrage? Wouldn't the government and society as a whole be trying to fix this? Wouldn't we be drafting class action lawsuits and demanding justice? Well, check your pockets because I'm guessing you have this drug on you right now and it's your smartphone.

When you think about addiction, what comes to mind? Substances like drugs and alcohol, maybe sex, gambling? Do you think about computer code, apps, algorithms? As a Child in Adolescent Psychiatry Fellow, I am here to advocate for my patients, who are mostly kids age 5 to 19, who I'm seeing become addicted to their phones in mass, and the purpose of this talk is to shift your perception of addiction because we must start viewing smartphones, technology and social media in the context of addiction, if we want to reverse our current mental health crisis.

What we know about addictions neurologically, is that all addictive things including your smartphone increase dopamine in your brain at the levels of the tech-mental, ventral-tegmental area and the nucleus-acumens. Dopamine makes you feel good; satisfied; but also increases your drive. That drive is important because it's the drive that makes you do things over and over again.

For example, in caveman times if you found a new food source that helped you survive, say a big bush of wild blueberries, your brain would release a tonne of dopamine to make you feel good about your find and increase that drive to do it again.

In today's world there are endless apps that are trying to trick your brain into releasing dopamine without being fundamentally good for you. App developers have found a way to hack a reward and addiction pathway and they did it without a substance; they did it with computer code. Addiction in 2022, has become nothing more than zeros and ones.

Typically, we define addiction by these signs. Play along and see how many you have. Like, hazardous use using while driving, or when it's dangerous. Impaired social functioning like isolating yourself or giving up on your social roles. Having withdrawal from your phone, cravings for your phone. Increased tolerance like your screen time creeping up. Trying to cut down on your use and being unable to. Spending more time using than you initially anticipated, and giving up your hobbies and interests to use.

How many of you have that oh crap that's me feeling? And if you don't see it in yourself how many people around you do you see these signs in? So if we're seeing everyone become addicted to their phone you'd think that the Mental Health Community would be doing something about this, right? But this addiction came on so fast that academia has yet to catch up and, as of now, none of the top mental health associations in the world have even recognized this as a real diagnosis the

DSM-5 the diagnostic and statistical manual psychiatry's Bible has no official diagnosis for phone addiction the ICD-10 and 11 the international billing and coding system we use in America.

Now there is a code for getting sucked up into a jet engine and requiring medical care but there's no code for phone addiction. If there's no code, it makes it incredibly difficult to diagnose bill for and track. We have to come to a consensus on a diagnosis and a code because right now, without an official diagnosis, the world has yet to even admit that this is a real problem and you don't have to be a psychiatrist to know that the first step in treating an addiction is admitting that you have a problem.

How pervasive is this issue? Well to compare it to smoking, at the peak of the tobacco industry in 1954 only 45 percent of Americans smoked. Right now 97% of Americans have a smartphone and surveys indicate about half of them feel addicted to them already.

Here's why your smartphone is a super addiction. It places multiple addictive elements into one device: video games, pornography, sports, gambling, shopping social media, instant messaging...By placing these all into one device that lives in your pocket, you're walking around with an addiction ticking timebomb that you have unlimited access to.

Your phone is programmed to buzz and ding you at just the right rate to make sure that you're constantly thinking about your phone. And your screen time creeps up. Now the phones are specifically designed to be addictive, but there are so many other reasons why this is a super addiction like rapid acceptance.

Remember phones smartphones have only been around for the past 15 years. The iPhone was released in 2007 and in 2022, 95% of American teens have a smartphone and use them on average up to nine hours per day. The original

telephone took over 100 years to democratize - where everyone has access to it. The smartphone did it and less than a fifth of the time.

Smartphones have revolutionized (a very sexy topic in 2022) supply chains. Most addictions have to be planted, grown, processed, shipped, packaged, prepared and consumed. Smartphones eliminate almost all of these steps and allow app developers to deliver free legal addictive content to anyone anywhere instantly.

Imagine if there was a physical drug that was free, legal, addictive and put right in your pocket whenever you craved it. How many people would be using it?

Smartphones will continue to evolve at the pace of the internet age: quickly and continuously. Look how fast Tick Tock was able to take over for an entire young generation. That's because it was specifically designed to be more addictive, designed to be harder to put down.

This will keep on happening where every few weeks, more addictive apps will come out to replace their watered-down predecessors. But perhaps the most important reason why smartphones are a super addiction is because it's socially acceptable; it's typically frowned upon to give your kids smokes booze...but giving your kids the new iPhone for Christmas makes you a good parent, right? At least that's what the commercials tell us.

I would argue that the phone is worse but society doesn't see it that way and that's why we've all been caught up in this current of addiction with little or no resistance from us. So I hope by now I've convinced you that your smartphone might be just a little bit addictive. The thing that worries me the most, as a mental health provider, is addictions have what we call co-morbidities or other mental health problems that go with the addiction: depression, anxiety, substance use, obsessive compulsive disorder... have all been studied and found linked to excessive smartphone use.

Not to mention the social problems that are coming as a result because I know lots of kids who get two to three messages per day telling them that they're worthless and that they should kill themselves. Kids face constant judgment from online and social media which I think might be making social anxiety worse. Sexual predation is rampant and psychologically scarring an entire generation. You want to know how bad this problem is? Ask any teenage girl how many times she's been pressured for nudes.

The longer we let smartphone addiction go on, the worse these issues will become, especially in this next generation who is growing up with an addiction. Because, when your brain develops with an addiction, the changes to brain physiology and personality can be dramatic and I'm not just talking about the very young. As brain development is complete in the early 20s for females and in the mid 20-s for males, addiction literally rewires your brain.

We've all heard the stories about someone addicted to methamphetamine or heroin who lies, cheats and steals from their family to support their drug addiction and ruins their relationships in the process. I've met young people who impulsively steal credit cards from anyone they can to support their online video game addiction similarly ruining their relationship and trust with their family.

When your brain develops with an addiction, everything else comes second. The rates of those comorbidities mentioned go up, academic performance goes down. Imagine how much creativity, empathy, productivity and connectivity we will lose if everyone in society is on their phone for nine hours every day.

Training as a child psychiatrist has taught me that in the very young brain development is dependent upon those daily interactions with caregivers. And babies these days are having to compete with smartphones for the attention of their parents. Let's face it, screens have fundamentally changed parenting directly. And how much kids are on their screens, and indirectly by how much their parents are on their

screens. I wonder as to how many collective hours of that brain development, of that interaction, we have lost because parents are spending hours a day on their phone just scrolling.

This is a topic of personal interest to me as I'm going to be a first-time father in 2023. Thank you. Right now, my screen time app tells me I'm on my phone from three to five hours per day and I'm curious as to how that will change me as a person and as a parent.

I think it's safe to say that cutting back on my phone use would give me more of that interaction time with my child, would make me a better parent but to be honest with you I don't know if I can because I've tried before and I failed. But I'm going to keep on trying because we all have a social obligation to make a world that we want.

If we want my son, your children, your grandchildren to grow up in a world where a kid can go back to being a kid, where two-year-olds aren't handed an addiction that consumes a third of their entire existence. A world with less depression, anxiety and suicide.

We have to take a hard look at our relationships with our phones because honestly and, you might not believe this, I'm not saying that smartphones are a bad thing. They're a wonderful tool in many cases and getting rid of them all together is just not realistic; I'm saying that we have to have a healthy balance with our phones before we fall too deeply into this addiction.

What do we do about it? There are four things you can do to reduce problematic use in young people. Number one, set screen time limiting apps and set a password. Kids aren't dumb they'll just turn them off and they do all the time.

Number two, take their phone away at night. Sleep is imperative to mental health and brain development and too many kids these days are staying up all night on their phone when they should be sleeping.

Number three, talk to your kids about this. Education is an important piece. If kids know it's addictive and it's not very good for them they're less likely to start.

And number four, take a look in the mirror. We have to set a good example for our young people. We cannot expect them to make a change if we ourselves are unwilling to make a change.

Those are some of the individual things you can do to stop the bleeding, if you will, on this addiction. The mental health community and the medical community have some work to do too. Like I mentioned earlier, we've got to come to a consensus on that code and the diagnosis. We have to start raising awareness about this. We have to start being diligent about asking our patients about their digital media consumption and screening for signs of addiction.

But perhaps the most important thing we can all do as a society is hold the tech companies accountable for what they're doing because if we don't they're just going to keep on doing it. The good news is we were able to hold the tobacco companies accountable and that helped turn the tide in the smoking epidemic. We can do the same thing now and I don't want to villainize the tech industry too much because I think there's a really wonderful opportunity for collaboration moving forward.

Imagine a world in which the tech industry shares their data with researchers so that we can understand this addiction and reverse it and what if we're also able to use that data to be able to identify people at the beginning stages of depression, anxiety, bipolar disorder, schizophrenia...to make sure that those people get the right help sooner. The tech companies would go from being a part of a problem to being a part of the broader solution. What a redemption.

Okay I know this is a little overwhelming and it's supposed to be. We're nowhere near those long-term, research driven solutions. Today's journey starts with raising awareness to an issue that's been right in front of our eyes for the past 15 years. That smartphone addiction is real. Turning the tide on this new pandemic starts right now.

[Applause]